Title:

Smart Urban-Mobility: Hourly Bicycle rental demand prediction using weather & temporal features.

Description:

The Seoul Bike Sharing Demand dataset, available from the UCI Machine Learning Repository, contains 8,760 hourly observations of public bicycle rentals in Seoul, South Korea, covering December 2017 to November 2018. It includes the number of bicycles rented per hour, date and time information, weather variables such as temperature, humidity, wind speed, rainfall and snowfall, and contextual features including season, holiday status and whether the bike-sharing system was operating.

What we are going to do:

Our project aims to develop a machine-learning system to predict hourly bicycle rental demand using the Seoul Bike Sharing Demand dataset from the UCI Machine Learning Repository. The dataset contains hourly rental counts along with weather and temporal features, including temperature, humidity, wind speed, rainfall, season, holidays and working-day status.

The project will begin with data exploration, cleaning and preprocessing to identify patterns and prepare suitable features for modelling. We will develop and compare a neural network model, a team-designed hybrid model and a transfer-learning model, as required by the project specification. Hyperparameter optimisation will be performed to improve model performance. The models will be evaluated using metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2, with chronological data splitting to prevent data leakage. Finally, we will develop an interactive web application to display hourly demand predictions, visualise demand patterns and compare model performance. The project aims to support better bicycle availability planning for urban bike-sharing services.

Data set Link:

https://archive.ics.uci.edu/dataset/560/seoul+bike+sharing+demand